

The expression of number and (in)definiteness in two varieties of Central Kurdish

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Abstract

In Central Kurdish, plural marking appears in different positions in indefinite, definite, and demonstrative noun phrases, and there are certain morpheme orders that cannot be derived by standard approaches to movement. To address this issue, we propose a reanalysis of number and definiteness morphology in this language, one that is based on a comparative investigation of the Silemānī and Ardaḡānī varieties. Building on insights by Wiltschko (2008, 2009), we represent number as a PLURALIZER (PL) whose adjunction site depends on whether DP projects. In indefinite noun phrases, PL modifies *n*, the head of a functional projection that dominates NP, and in definite and demonstrative noun phrases, PL modifies D. We then draw a connection to research on strong versus weak definiteness and decompose the definite suffix *-aká* into *-ak*, a component that expresses anaphoricity, and *-á*, a component that expresses uniqueness. Finally, we argue that PL and D undergo fusion in anaphoric constructions, in which case these nodes are realized by the same form, *-ān*, in both Silemānī and Ardaḡānī Kurdish.

Keywords: number, definiteness, demonstratives, anaphoricity, uniqueness, nominal linkers, roll-up movement, Kurdish, Iranian languages

1. Introduction

A fundamental concern in work on Central Kurdish is how to account for the variable position of number morphology (Karimi 2007, Tahir 2018, Haig 2019a, Kalami 2020).¹ Assuming that the base order of the noun phrase is D AP Num N, it is not immediately evident how to capture the data in (1), where the modifier is adjacent to the noun but the plural suffix is not.²

- (1) a. *zimānnās-a ganj-ak-ān* N AP D Num
linguist-IZ young-DEF-PL
'the young linguists'
- b. *aw zimānnās-a ganj-ak-ān* N AP D Num
DIST linguist-IZ young-DEF-PL
'those young linguists'

Movement of N alone cannot derive N AP D Num from D AP Num N, nor can pied-piping involving N. In this chapter, we analyze number morphology in Central Kurdish as a PLURALIZER (PL) that is sensitive to the (in)definiteness of the noun phrase. In structures that lack DP, PL

¹There are approximately 8,000,000 speakers of Central, or Sorānī, Kurdish: 4,750,000 in the Kurdistan Region of northern Iraq, 3,000,000 in the provinces of Kermanshah, Kurdistan, and West Azerbaijan in northwestern Iran, and 250,000 in diasporic communities, the majority of which are located in Europe and North America (Sheyholislami 2021).

²We largely adhere to Thackston's (2006) system when transliterating Central Kurdish but use the macron rather than the circumflex to distinguish the "long" vowels *ā*, *ī*, and *ū* from their "short" counterparts *a*, *i*, and *u*.

adjoins to *n*, the head of an intermediate functional projection that hosts the indefinite suffix *-ək*, and in structures that contain DP, PL adjoins to D. As for the definite suffix *-aká*, which we take to be bimorphemic, we suggest that *-ak* encodes anaphoricity, or strong definiteness, and that *-á* encodes uniqueness, or weak definiteness. Moreover, we claim that plurality and uniqueness, the latter of which includes maximality, are both expounded by the suffix *-ān* in anaphoric noun phrases. According to the proposal that we develop, PL modifies D in all definite expressions and is spelled out together with D in anaphoric ones.

In section 2, we present the morpheme order of indefinite, definite, and demonstrative noun phrases in Central Kurdish and identify the major differences between the Silemānī and Ardaḡānī varieties. In section 3, we argue that roll-up movement takes place in this language based on the rigid ordering of direct and indirect modifiers and of comparative adjectives. In section 4, we raise the possibility that number is a modifier of other heads in Central Kurdish and demonstrate how this approach works in indefinite constructions. In section 5, we decompose the “singular” definite suffix *-aká* into *-ak* and *-á* and the “plural” definite suffix into *-ak* and *-ān*, where *-ak* conveys anaphoricity, *-á* conveys maximality, and *-ān* conveys plurality as well as maximality. In section 6, we formalize the remaining details of our account and provide structures for all definite and demonstrative constructions. In section 7, we conclude with a review of our findings.

2. Number and (in)definiteness in Central Kurdish

The Silemānī variety of Central Kurdish is spoken in Silemānī (Sulaymaniyah), Iraq and serves as the basis for the written standard, and the Ardaḡānī variety is spoken in Sina (Sanandaj), Iran (MacKenzie 1961; Abdulla & McCarus 1967; McCarus 1997; Blau 2000; Thackston 2006; McCarus 2007, 2009; Haig 2019b; Matras 2019; Kalami 2020; Sheyholislami 2021; Kalami 2023). Ardaḡānī Kurdish exhibits two morphological distinctions that are not found in Silemānī Kurdish, one that relates to the form of plural marking and another that relates to the function of demonstrative expressions. In sections 2.1 through 2.3, we describe the facts that inform our reanalysis of number and definiteness morphology, and in section 2.4, we summarize the morpheme order of the noun phrase in both varieties under study.

2.1. Indefinite noun phrases

In indefinite constructions, such as those in (2) and (3), the plural suffix takes a different form in Silemānī versus Ardaḡānī Kurdish, while the indefinite suffix takes the same form (MacKenzie 1961; Abdulla & McCarus 1967; McCarus 1997; Blau 2000; Thackston 2006; McCarus 2007, 2009; Öpengin 2016; Tahir 2018; Haig 2019b; Kalami 2020; Jambrović & Hassan 2023; Kalami 2023; Karim 2023).³

- | | | | |
|-----|----|--|----------|
| (2) | a. | <i>zimānnās-ān</i>
linguist-PL
‘linguists’ | Silemānī |
|-----|----|--|----------|

³Stress is typically word final in Central Kurdish (McCarus 1997; Blau 2000; Thackston 2006; McCarus 2007, 2009; Öpengin 2016). Affixes that are exceptionally stressed are marked with an acute accent (e.g., the progressive prefix *á-* or *dá-*), and those that are exceptionally unstressed are marked with a grave accent (e.g., the indefinite suffix *-ək*). Like Thackston (2006), we use an acute accent to differentiate the stressed definite suffix *-á* from the unstressed linker *-a*.

- b. *zimānnās-èk*
linguist-INDF
'a linguist'
- (3) a. *zabānshnās-gal* Ardaḡānī
linguist-PL
'linguists'
- b. *zabānshnās-èk*
linguist-INDF
'a linguist'

The etymological source of all three elements is well documented: *-ān* is derived from the plural oblique case ending *-ān*, *-gal* is derived from the noun *gal* 'flock, herd', and *-èk* is derived from the numeral *yek* 'one' (Mahmoodi-Bakhtiari & Rezai-Baghbidi 2019, Taheri 2021, Karim 2023). When the plural suffix and the indefinite suffix are used together, as in (4), the plural suffix precedes the indefinite suffix (Tahir 2018: 97).⁴

- (4) *kas-ān-èk han zor hal parist=in.* Bingird
person-PL-INDF have.3PL very opportunity worshipper=are.3PL
'There are (some) people who are very opportunist.'

Although these suffixes rarely cooccur, we deliberately include both of them in our examples to illustrate their position with respect to other elements in the noun phrase.

Modifiers follow number and indefiniteness morphology in Central Kurdish and appear with an unstressed segment that is known as *īzāfa* and attaches to the preceding constituent (MacKenzie 1961, Abdulla & McCarus 1967, Blau 2000, Thackston 2006, Karimi 2007, McCarus 2009, Öpengin 2016, Tahir 2018, Haig 2019b, Kalami 2020, Jambrović & Hassan 2023, Kalami 2023, Karim 2023).⁵ In the indefinite expressions in (5) and (6), the linker is always pronounced in the Silemānī variety but is only pronounced after vowel-final stems in the Ardaḡānī variety.

- (5) a. *zimānnās[-ī ganj]* Silemānī
linguist-IZ young
'young linguist'
- b. *nānawā[-y ganj]*
baker-IZ young
'young baker'
- c. *zimānnās-ān-èk[-ī ganj]*
linguist-PL-INDF-IZ young
'(some) young linguists'

⁴Here and elsewhere in the chapter, we adapt the transliterations of Central Kurdish data from other sources to reflect Thackston's (2006) notation.

⁵See Tahir 2018 for arguments that *īzāfa* is a clitic rather than a suffix. We do not take a stance regarding this issue and use the neutral term "linker".

- (6) a. *zabānshnās*[-Ø *jiwān*] Ardafānī
 linguist-IZ young
 ‘young linguist’
- b. *nānwā*[-y *jiwān*]
 baker-IZ young
 ‘young baker’
- c. *zabānshnās-gal-èk*[-Ø *jiwān*]
 linguist-PL-INDF-IZ young
 ‘(some) young linguists’

Kalami (2020) observes that the indefinite suffix can also follow modifiers in Ardafānī Kurdish, as in (7), but notes that this morpheme order is less common than the one in (6c).

- (7) *zabānshnās-gal*[-Ø *jiwān*]-èk Ardafānī
 linguist-PL-IZ young-INDF
 ‘(some) young linguists’

One could paraphrase *zabānshnāsgalèk jiwān* ‘(some) young linguists’ in (6c) as ‘(some) linguists, who are young’ because the indefinite suffix only takes scope over the noun in this case. This possibility does not extend to *zabānshnāsgal jiwānèk* ‘(some) young linguists’ in (7), however, where the indefinite suffix takes scope over the adjective as well.

2.2. Definite noun phrases

The traditional view of definite constructions in Central Kurdish, such as those in (8) and (9), is that the definite suffix precedes the plural suffix (MacKenzie 1961; Abdulla & McCarus 1967; Blau 2000; Thackston 2006; Karimi 2007; McCarus 2007, 2009; Öpengin 2016; Tahir 2018; Haig 2019a,b; Kalami 2020; Jambrović & Hassan 2023; Kalami 2023; Karim 2023).

- (8) a. *zimānnās-aká* Silemānī
 linguist-DEF
 ‘the linguist’
- b. *zimānnās-ak-ān*
 linguist-DEF-PL
 ‘the linguists’
- (9) a. *zabānshnās-aká* Ardafānī
 linguist-DEF
 ‘the linguist’
- b. *zabānshnās-ak-ān*
 linguist-DEF-PL
 ‘the linguists’

In this case, the plural suffix has the same form in Silemānī and Ardafānī Kurdish, and the data in (8b) and (9b) raise the question of whether the stressed *-á* of the definite suffix is elided before the *-ā* of the plural suffix or simply not present to begin with.

In both varieties, modifiers occur inside of definiteness and number morphology and are preceded by an unstressed *-a* that is silent after certain vowel-final stems, including those ending in *-a* and *-ā* (MacKenzie 1961; Abdulla & McCarus 1967; Blau 2000; Thackston 2006; Karimi 2007; McCarus 2007, 2009; Öpengin 2016; Tahir 2018; Haig 2019a,b; Kalami 2020; Jambrović & Hassan 2023; Kalami 2023; Karim 2023). We gloss this vowel as *izāfā* in (10) and (11) and assume that its realization as *-a* instead of *-ī* or *-y* is a reflex of definiteness concord (Karimi 2007, Salehi 2018, Tahir 2018, Kalami 2020).

- (10) a. *zimānnās[-a ganj]-ak-ān* Silemānī
 linguist-IZ young-DEF-PL
 ‘the young linguists’
- b. *nānawā[-∅ ganj]-ak-ān*
 baker-IZ young-DEF-PL
 ‘the young bakers’
- (11) a. *zabānsbnās[-a jiwān]-ak-ān* Ardatānī
 linguist-IZ young-DEF-PL
 ‘the young linguists’
- b. *nānwā[-∅ jiwān]-ak-ān*
 baker-IZ young-DEF-PL
 ‘the young bakers’

Some authors consider the *-a* linker that appears in definite expressions to be the same *-a* that is found in many compounds (MacKenzie 1961, Blau 2000, Thackston 2006, McCarus 2009, Öpengin 2016). The example in (12) is particularly informative because the presence of the indefinite suffix confirms that this occurrence of *-a* is unrelated to definiteness (MacKenzie 1961: 65).

- (12) *kīlk-a-kar-èk* Silemānī
 tail-*a*-donkey-INDF
 ‘a donkey’s tail’

In previous work, we explored the viability of a compounding-based analysis of definite noun phrases, and since then, we have identified more phenomena that undermine this possibility (Jambrović & Kahnemuyipour 2023). For instance, even the syntactically complex APs in (13) and (14) appear inside of definiteness and number marking.

- (13) a. *zimānnās[-a zor ganj]-ak-ān* Silemānī
 linguist-IZ very young-DEF-PL
 ‘the very young linguists’
- b. *pardākb[-a pīř la āw]-ak-ān*
 glass-IZ full of water-DEF-PL
 ‘the glasses full of water’

- (14) a. *zabānshnās[-a fira jiwān]-ak-ān* Ardaḩānī
 linguist-IZ very young-DEF-PL
 ‘the very young linguists’
- b. *līwān[-a pīř la āw]-ak-ān*
 glass-IZ full of water-DEF-PL
 ‘the glasses full of water’

Crosslinguistically, genuine compounds do not permit modifiers of modifiers, as shown by *black-board* versus **veryblackboard* in English, so we pursue a different approach to the morpheme order of definite constructions in Central Kurdish.

2.3. Demonstrative noun phrases

We begin the discussion of demonstrative expressions with Ardaḩānī Kurdish, where two patterns are attested. According to Kalami (2020, 2023), deictic uses have the form in (15), where the proximal morpheme *am* ‘this, these’ and the distal morpheme *aw* ‘that, those’ are both possible, and anaphoric uses have the form in (16), where only *aw* is possible.⁶

- (15) a. {*am* / *aw*} *zabānshnās-á* Ardaḩānī
 PROX DIST linguist-DEF
 ‘this/that linguist’ (deictic)
- b. {*am* / *aw*} *zabānshnās-gal-á*
 PROX DIST linguist-PL-DEF
 ‘these/those linguists’ (deictic)
- (16) a. {**am* / *aw*} *zabānshnās-aká* Ardaḩānī
 PROX DIST linguist-DEF
 ‘that linguist’ (anaphoric)
- b. {**am* / *aw*} *zabānshnās-ak-ān*
 PROX DIST linguist-DEF-PL
 ‘those linguists’ (anaphoric)

The constructions in (15) can, but need not, be accompanied by a gesture that indicates the referent, suggesting that they are not exclusively deictic in nature. For this reason, we propose to reclassify them on the basis of uniqueness rather than deixis in accordance with Schwarz’s (2009, 2013, 2019) account of languages that morphologically distinguish uniqueness from anaphoricity. In broad terms, uniqueness conveys that there is a unique maximal object or sum of objects that has a given property, and anaphoricity conveys that the referent of an expression has been previously introduced in the discourse. For example, the demonstrative noun phrase in (17a) can be used to introduce a new referent, one that is known to the speaker but not yet to the

⁶Kalami builds on work by Taghipour (2019, 2021), who first described a contrast between deictic and anaphoric definite noun phrases in Laki, a closely related Kurdish language.

addressee, whereas the demonstrative noun phrase in (17b) implies that the referent is already known to the addressee through prior mention.⁷

- (17) a. {*am* / *aw*} *zabānshnās-gal-á wā dweka dī=yān=im* Ardaḩānī
 PROX DIST linguist-PL-DEF that yesterday saw=3PL=1SG
 ‘these/those linguists whom I met yesterday’ (unique)
- b. *aw zabānshnās-ak-ān wā dweka dī=yān=im*
 DIST linguist-DEF-PL that yesterday saw=3PL=1SG
 ‘those linguists whom I met yesterday’ (anaphoric)

In section 5, we argue that *-ak* encodes anaphoricity in both varieties of Central Kurdish and situate this element in a higher level of the noun phrase than *-á/-ān*.

Turning to Silemānī Kurdish, the sole demonstrative construction is that in (18), which has the same morpheme order as unique demonstratives in Ardaḩānī Kurdish (MacKenzie 1961; Abdulla & McCarus 1967; Blau 2000; Thackston 2006; McCarus 2007, 2009; Tāhir 2018; Haig 2019b; Jambrović & Hassan 2023; Karim 2023).

- (18) a. {*am* / *aw*} *zimānnās-á* Silemānī
 PROX DIST linguist-DEF
 ‘this/that linguist’
- b. {*am* / *aw*} *zimānnās-ān-á*
 PROX DIST linguist-PL-DEF
 ‘these/those linguists’

As one might expect, these expressions can be used for unique as well as anaphoric reference since there is no competing form that includes *-ak*.

Independently of the variety under consideration, modifiers have the same distribution in demonstrative noun phrases that they do in other definite noun phrases. As shown in (19) and (20), modifiers are preceded by a linker of the form *-a* and occur inside of number and definiteness morphology (MacKenzie 1961; Abdulla & McCarus 1967; Blau 2000; Thackston 2006; McCarus 2007, 2009; Öpengin 2016; Tāhir 2018; Haig 2019a,b; Kalami 2020; Jambrović & Hassan 2023; Kalami 2023; Karim 2023).

- (19) a. *am zimānnās[-a ganj]-ān-á* Silemānī
 PROX linguist-IZ young-PL-DEF
 ‘these young linguists’
- b. *aw zimānnās[-a ganj]-ān-á*
 DIST linguist-IZ young-PL-DEF
 ‘those young linguists’
- (20) a. *am zabānshnās[-a jiwān]-gal-á* Ardaḩānī
 PROX linguist-IZ young-PL-DEF
 ‘these young linguists’ (unique)

⁷An anonymous reviewer drew our attention to the fact that both types of demonstrative constructions can be followed by a restrictive relative clause in Ardaḩānī Kurdish, and we are grateful to Mojgan Osmani (p.c.) for highlighting the semantic differences between them.

- b. *aw zabānshnās[-a jiwān]-ak-ān*
DIST linguist-IZ young-DEF-PL
‘those young linguists’ (anaphoric)

Here as well, we take the *-a* linker to be a definiteness-conditioned allomorph of *izāfa*, which is otherwise spelled out as *-ī* or *-y*.

2.4. Summary

In Tables 1 and 2, we provide the type, form, and morpheme order of the noun phrase in Silemānī and Ardaḡānī Kurdish according to the discussion in sections 2.1 through 2.3.

Table 1: The noun phrase in Silemānī Kurdish

Indefinite	N- <i>ān-èk</i> [- <i>ī</i> AP]	N Num Indf AP
	N-PL-INDF-IZ AP	
Definite	N[- <i>a</i> AP]- <i>ak-ān</i>	N AP D Num
	N-IZ AP-DEF-PL	
Demonstrative	<i>aw</i> N[- <i>a</i> AP]- <i>ān-á</i>	Dem N AP Num D
	DIST N-IZ AP-PL-DEF	

Table 2: The noun phrase in Ardaḡānī Kurdish

Indefinite (more common)	N- <i>gal-èk</i> [- $\emptyset$ AP]	N Num Indf AP
	N-PL-INDF-IZ AP	
Indefinite (less common)	N- <i>gal</i> [- $\emptyset$ AP]- <i>èk</i>	N Num AP Indf
	N-PL-IZ AP-INDF	
Definite	N[- <i>a</i> AP]- <i>ak-ān</i>	N AP D Num
	N-IZ AP-DEF-PL	
Demonstrative (unique)	<i>aw</i> N[- <i>a</i> AP]- <i>gal-á</i>	Dem N AP Num D
	DIST N-IZ AP-PL-DEF	
Demonstrative (anaphoric)	<i>aw</i> N[- <i>a</i> AP]- <i>ak-ān</i>	Dem N AP D Num
	DIST N-IZ AP-DEF-PL	

It is uniformly the case that modifiers follow number morphology in indefinite constructions but precede number morphology in definite and demonstrative constructions. Furthermore, there seems to be an interaction between number and definiteness in Ardaḡānī Kurdish given that the plural suffix is only realized as *-ān* when it follows the definite suffix *-ak*.

3. Modifiers and roll-up movement

In Central Kurdish, as in many Western Iranian languages, *izāfa* is only found with postnominal modifiers (MacKenzie 1961; Abdulla & McCarus 1967; Blau 2000; Thackston 2006; McCarus 2007, 2009; Öpengin 2016; Tahir 2018; Haig 2019b; Jambrović & Hassan 2023; Kalami 2023).

In (21), for example, the numeral *se* ‘three’ and the superlative adjective *bātābarztirīn* ‘tallest’ are both prenominal and reject *izāfa*, while the positive adjective *ganj* ‘young’ is postnominal and requires *izāfa*.

- (21) a. *se-(*)y zimānnās-*(i) ganj* Silemānī
 three-IZ linguist-IZ young
 ‘four young linguists’
- b. *bātābarz-tir-īn-*(i) zimānnās-*(i) ganj*
 tall-CMPR-SUP-IZ linguist-IZ young
 ‘the tallest young linguist’

This numeral, which ends in a vowel, exhibits the same behaviour in the Ardaḩānī equivalent of (21a) in (22).

- (22) *se-(*)y nānwā-*(y) jiwān* Ardaḩānī
 three-IZ baker-IZ young
 ‘three young bakers’

We now demonstrate that modifiers are subject to roll-up movement in Central Kurdish through an exploration of two types of ordering restrictions.

For Cinque (2010), “indirect” modifiers have stage-level, restrictive, and intersective interpretations, whereas “direct” modifiers have individual-level, nonrestrictive, and nonintersective interpretations. As illustrated in (23) and (24), indirect modifiers like *zīrak* ‘clever’ must appear further from the noun than direct modifiers like *tīorī* ‘theoretical’ in indefinite as well as definite expressions (Jambrović & Hassan 2023).

- (23) a. *zimānnās-ān-ī {tīorī-y zīrak / #zīrak-ī tīorī}* Silemānī
 linguist-PL-IZ theoretical-IZ clever clever-IZ theoretical
 ‘clever theoretical linguists’
- b. *zimānnās-a {tīorī-a zīrak / #zīrak-a tīorī}-ak-ān*
 linguist-IZ theoretical-IZ clever clever-IZ theoretical-DEF-PL
 ‘the clever theoretical linguists’
- (24) a. *zabānshnās-gal-Ø {tīorī-y zīrak / #zīrak-Ø tīorī}* Ardaḩānī
 linguist-PL-IZ theoretical-IZ clever clever-IZ theoretical
 ‘clever theoretical linguists’
- b. *zabānshnās-a {tīorī-a zīrak / #zīrak-a tīorī}-ak-ān*
 linguist-IZ theoretical-IZ clever clever-IZ theoretical-DEF-PL
 ‘the clever theoretical linguists’

If, as Cinque posits, the domain of indirect modification is structurally higher than the domain of direct modification, the word order in (23) and (24) can only be derived if the noun “collects” the direct modifier along its way to a landing site above the indirect modifier. This phenomenon has come to be known by various names in the literature, including pied-piping, roll-up movement, and snowballing (Sichel 2002, Aboh 2004, Shlonsky 2004, Cinque 2005, Laenzlinger 2005,

Cinque 2010, Kahnemuyipour 2014, Atlamaz 2016). In work on Central Kurdish, Tahir (2018) claims that roll-up movement is limited to definite constructions, but the rigid ordering of the indirect modifier after the direct modifier in the indefinite constructions in (23a) and (24a) renders this view untenable.

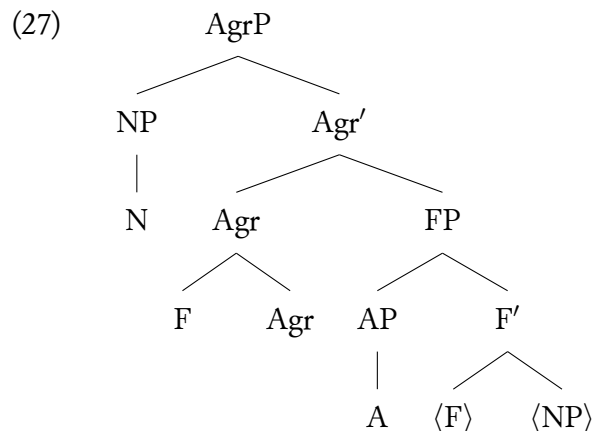
The next constraint involves comparative adjectives, which are derived by the suffix *-tir* in Central Kurdish. In (25) and (26), we show that the comparative adjective *bātābarztir* ‘taller’ necessarily follows the positive adjective *ganj/jiwān* ‘young’ (Jambrović & Hassan 2023).⁸

- (25) a. *zimānnās-ān-ī {ganj-ī bātābarz-tir / #bātābarz-tir-ī ganj}* Silemānī
 linguist-PL-IZ young-IZ tall-CMPR tall-CMPR-IZ young
 ‘taller young linguists’
- b. *zimānnās-a {ganj-a bātābarz-tir / #bātābarz-tir-a ganj}-ak-ān*
 linguist-IZ young-IZ tall-CMPR tall-CMPR-IZ young
 ‘the taller young linguists’
- (26) a. *zabānsbnās-gal-Ø {jiwān-Ø bātābarz-tir / #bātābarz-tir-Ø jiwān}* Ardaḷānī
 linguist-PL-IZ young-IZ tall-CMPR tall-CMPR-IZ young
 ‘taller young linguists’
- b. *zabānsbnās-a {jiwān-a bātābarz-tir / #bātābarz-tir-a jiwān}-ak-ān*
 linguist-IZ young-IZ tall-CMPR tall-CMPR-IZ young
 ‘the taller young linguists’

From a semantic perspective, *zimānnāsānī ganjī bātābarztir* ‘taller young linguists’ in (25a) and *zabānsbnāsgal jiwān bātābarztir* ‘taller young linguists’ in (26a) do not merely denote a set of individuals that have the properties Linguist, Young, and Taller. Rather, they express that there are two sets of individuals that have the properties Linguist, Young, and Tall and that the members of one set differ from those of the other in terms of their relative degree of tallness. Since these facts indicate that comparative adjectives must occupy a position from which they can take scope over positive adjectives, they reveal a flaw in Larson & Yamakido’s (2008) DP-shell approach to modification, where D is the head of the noun phrase, APs are complements of D, and NPs merge as specifiers of DP. In such a system, the only way to derive the word order in (25) and (26) would be for the positive adjective to c-command the comparative adjective, but this structure would not yield the correct interpretation. An account based on roll-up movement does not face this problem because the hierarchical relation between two or more modifiers is the inverse of their linear order.

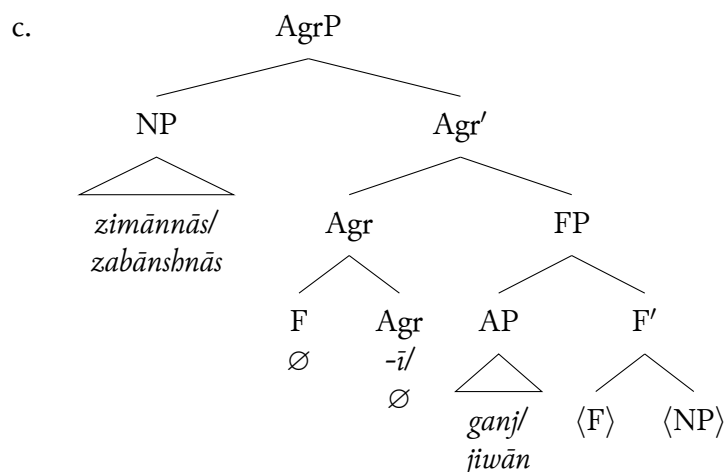
Throughout this chapter, we assume Shlonsky (2004) and Kahnemuyipour’s (2014) implementation of roll-up movement, where APs merge as specifiers of a functional phrase (FP) that is dominated by an agreement phrase (AgrP). As schematized in (27), F raises to form a complex head with Agr, and the NP raises to the specifier of AgrP.

⁸Kalami (2023) reports that comparative adjectives have flexible orderings in Ardaḷānī Kurdish but does not consider the difference in interpretation that results from positioning a comparative adjective closer to the noun than a positive adjective, nor the distinct prosody that may be needed to do so. For instance, *#tall younger linguists* in English can be improved by stressing the positive adjective, as in *TALL younger linguists*, the effect of which is contrastive focus.



Kahnemuyipour argues that Persian *ezāfe* is the realization of Agr, an analysis that we extend to Central Kurdish *izāfa*.⁹ For example, the indefinite noun phrases in (28a) and (28b) both have the structure in (28c).

- (28) a. *zimānnās-ī ganj* Silemānī
 linguist-IZ young
 ‘young linguists’
- b. *zabānsbnās-Ø jiwān* Ardaḡānī
 linguist-IZ young
 ‘young linguists’

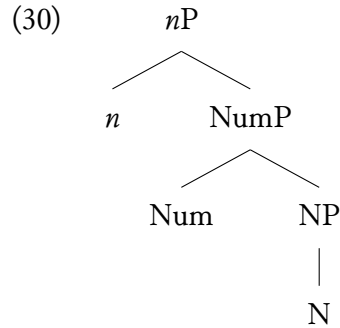


Considering the need for roll-up movement, it is not immediately apparent how to represent the more complex indefinite expressions in (29), where the indefinite suffix intervenes between the plural suffix and the modifier.

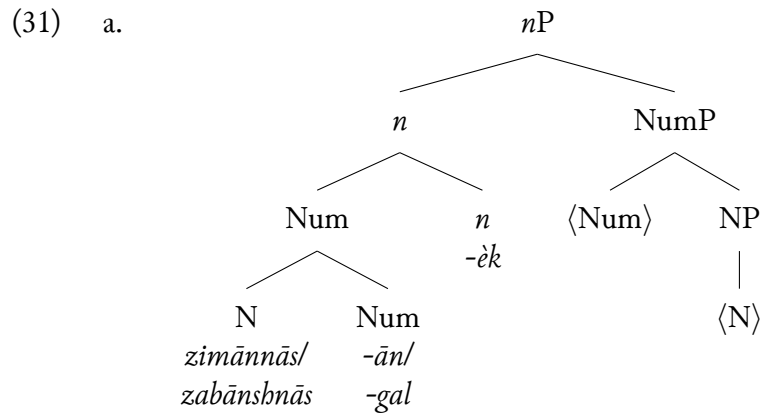
⁹In support of this view, there are many Northwestern Iranian languages where nominal linkers inflect for gender, number, and case (Holmberg & Odden 2008, Toosarvandani & van Urk 2014, Tahir 2018, Kahnemuyipour & Peters 2019, Gündoğdu et al. 2020, Gündoğdu 2023, Gündoğdu & Bulan 2023, Gündoğdu & Erbası 2023, Peters et al. 2023, Gündoğdu et al. 2024).

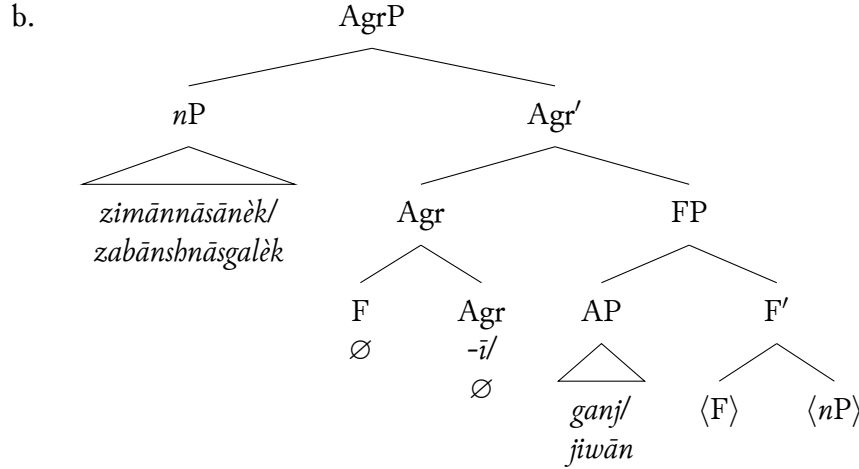
- (29) a. *zimānnās-ān-èk-ī ganj* Silemānī
 linguist-PL-INDF-IZ young
 ‘(some) young linguists’
- b. *zabānshnās-gal-èk-Ø jiwān* Ardaḡānī
 linguist-PL-INDF-IZ young
 ‘(some) young linguists’

One possibility is to adopt the structure that Julien (2003, 2005) proposes for the lower layers of the noun phrase in (30), where *nP* is a functional projection that serves as a secondary locus of definiteness.



The morpheme order of *zimānnāsānèkī ganj/zabānshnāsgalèkī jiwān* ‘(some) young linguists’ in (29) could then be attributed to the head movement in (31a) followed by the phrasal movement in (31b).





As we alluded to in section 1, it is not possible to maintain roll-up movement and capture the morpheme order in definite or demonstrative constructions, such as those in (32) and (33), where the plural suffix is not adjacent to the noun.

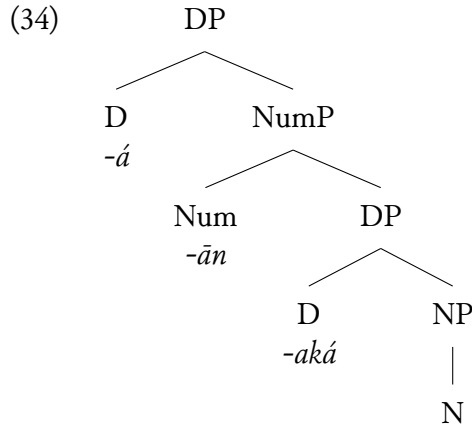
- (32) a. *zimānnās-a ganj-ak-ān* Silemānī
 linguist-IZ young-DEF-PL
 ‘the young linguists’
- b. *aw zimānnās-a ganj-ān-á*
 DIST linguist-IZ young-PL-DEF
 ‘those young linguists’
- (33) a. *zabānshnās-a jiwān-ak-ān* Ardaḡānī
 linguist-IZ young-DEF-PL
 ‘the young linguists’
- b. *aw zabānshnās-a jiwān-gal-á*
 DIST linguist-IZ young-PL-DEF
 ‘those young linguists’ (unique)
- c. *aw zabānshnās-a jiwān-ak-ān*
 DIST linguist-IZ young-DEF-PL
 ‘those young linguists’ (anaphoric)

In the solution that we develop, we argue that number enters the derivation at a higher level of the noun phrase in definite and demonstrative expressions than it does in indefinite expressions.

4. Number as an adjunct

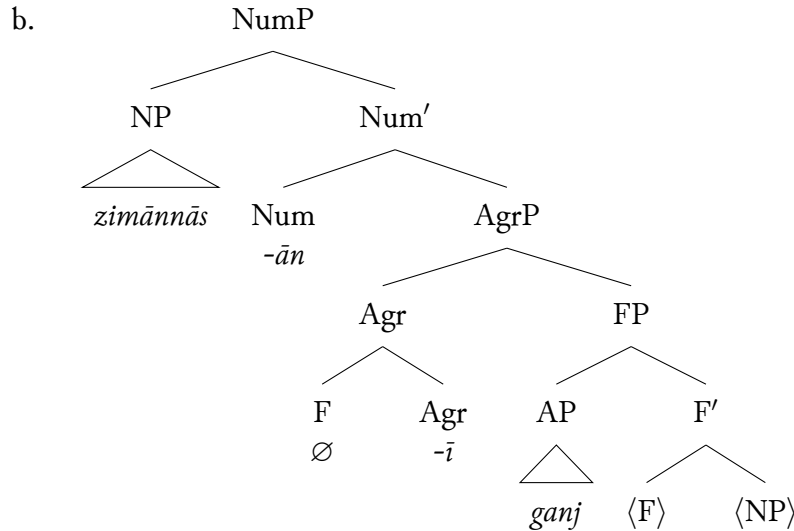
We begin this section by discussing two previous accounts of number morphology in Central Kurdish and the various challenges that they face. The first is by Tahir (2018), who proposes

that NumP occupies a position between two DP layers, as in (34), where each D is realized by a unique definite suffix.¹⁰



We find there to be three issues with this structure, the first of which is the redundancy that stems from assigning the stressed element *-á* to more than one head in the noun phrase. Second, the configuration in (34) is at odds with the established view in the semantics literature that pluralization precedes determination (Sharvy 1980; Link 1983, Krifka 1989; Landman 1989; Chierchia 1998a,b). Third, to derive the adjacency of the noun and the plural suffix in indefinite expressions, such as that in (35a), Tahir must limit movement to the NP itself, as shown in (35b).

- (35) a. *zimānnās-ān-ī ganj* Silemānī
 linguist-PL-IZ young
 ‘young linguists’



However, we have already demonstrated in section 3 that modifiers are subject to ordering restrictions in all types of noun phrases. If, as we maintain, the indefinite expressions in (36)

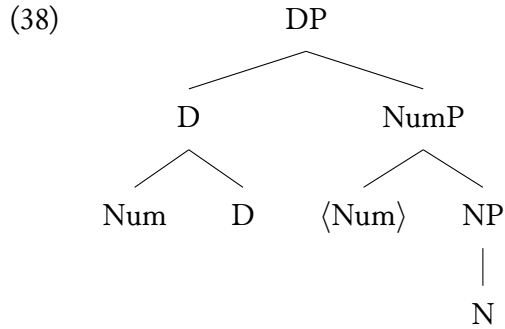
¹⁰In their analysis of Hawrāmī, another Northwestern Iranian language, Holmberg & Odden (2008) also consider NumP to dominate DP.

and (37) are derived through roll-up movement, just as their definite counterparts are, it must be the case that the plural suffix is generated in a position below the modifier.

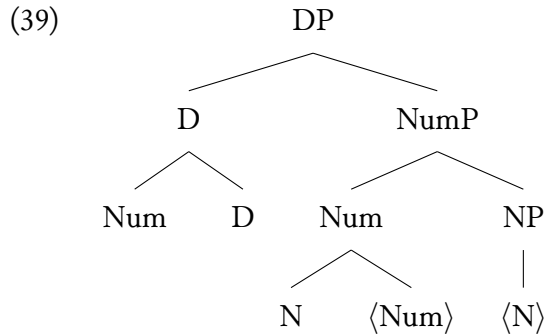
- (36) a. *zimānnās-ān-ī tīorī-y zīrak* Silemānī
 linguist-PL-IZ theoretical-IZ clever
 ‘clever theoretical linguists’
- b. *zimānnās-ān-ī ganj-ī bātābarz-tir*
 linguist-PL-IZ young-IZ tall-CMPR
 ‘taller young linguists’
- (37) a. *zabānshnās-gal-Ø tīorī-y zīrak* Ardaḡānī
 linguist-PL-IZ theoretical-IZ clever
 ‘clever theoretical linguists’
- b. *zabānshnās-gal-Ø jiwān-Ø bātābarz-tir*
 linguist-PL-IZ young-IZ tall-CMPR
 ‘taller young linguists’

In short, the approach to the noun phrase in (34) above raises theoretical and empirical concerns that are not readily dismissed.

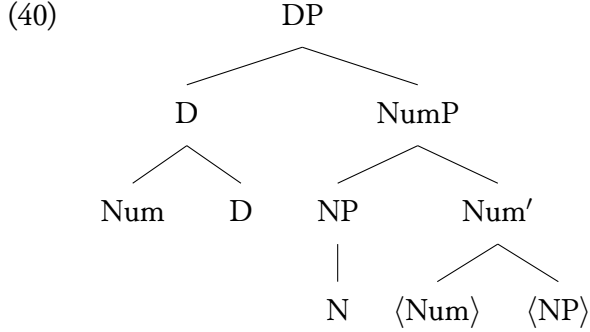
The second analysis, which was first proposed by Taghipour (2019, 2021) for Laki and later adopted by Kalami (2020) for Ardaḡānī Kurdish, implements Num-to-D raising in definite constructions and is illustrated in (38).



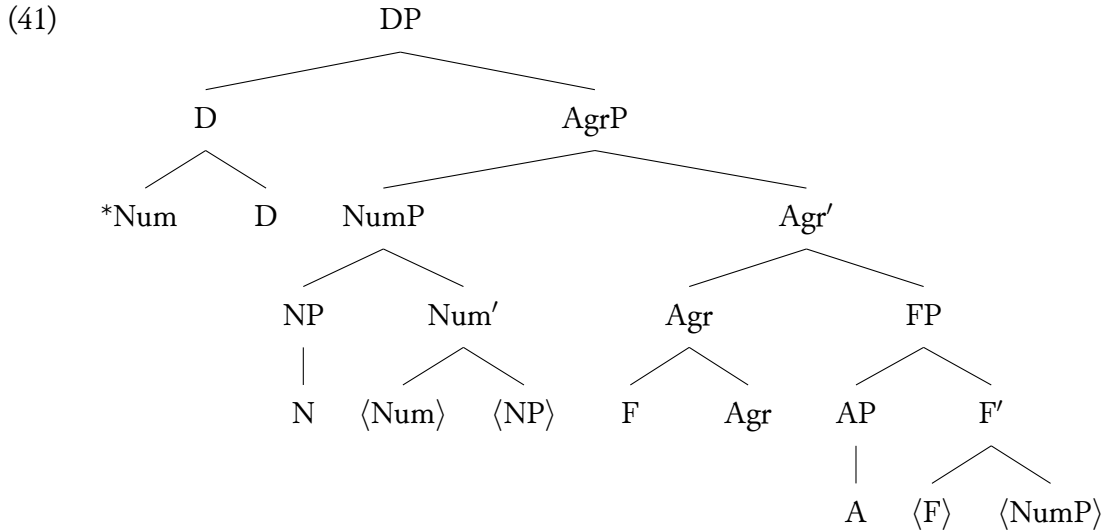
This move has consequences that are not easily remedied either. Unless one stipulates that Num can be excorporated from the complex head in (39), which is generally regarded to be an illicit type of movement, Num-to-D raising is mutually exclusive with N-to-Num raising (Roberts 1991, Matushansky 2006, Roberts 2010, Dékány 2018, Hsu 2021).



To capture the adjacency of the noun and the plural suffix in certain expressions, Kalami instead resorts to phrasal movement of NP to the specifier of NumP, as in (40), but raising the complement of a head to the specifier of the same phrase violates antilocality (Abels 2003, Grohmann 2003, Bošković 2005, Boeckx 2007).



Independently of this particular issue, Num-to-D movement seems to be incompatible with roll-up movement in that Num cannot escape the specifier of AgrP to raise to D in (41).¹¹



The question, therefore, is how to explain the adjacency of number and definiteness morphology in Central Kurdish without relying on head movement.

According to Wiltschko (2008, 2009), there are languages where plural marking is neither inflectional nor derivational and instead behaves as an optional modifier. One reason to suspect that the status of number is different in Central Kurdish is that the plural suffix is not needed in (42) for the noun *mishik* ‘mouse’ to denote different mice.

- (42) a. *tawāw-ī rozh mishik dá-bāt zhūr-aká-wa.* Silemānī
 whole-IZ day mouse DUR-came.3SG room-DEF-in
 ‘The whole day mice kept coming in the room.’

¹¹One could, in principle, avoid this problem by stipulating that roll-up movement is delayed until D merges, thereby allowing Num-to-D raising to take place first, but we do not consider there to be sufficient rationale for such a concession to the theory.

- b. *tawāw-Ø rozh mishik á-bāta naw otāq-aká.* Ardaḡānī
 whole-IZ day mouse DUR-came.3SG in room-DEF
 ‘The whole day mice kept coming in the room.’

In Hindi, another Indo-Iranian language that has number morphology and yet permits bare singular nouns, only the plural noun *cuuhee* ‘mice’ in (43a) can be used in this type of context (Dayal 2004: 404).

- (43) a. *puure din kamre meN cuuhee ghuste rabee.* Hindi
 whole day room in mice kept coming.in
 ‘Mice kept coming into the room the whole day.’
 b. # *puure din kamre meN cuubaa ghustaa rahaa.*
 whole day room in mouse kept coming.in
 ‘The same mouse kept coming into the room the whole day.’

The contrast between these languages is found with human-denoting nouns as well. As shown in (44) and (45), the bare singular noun *mindāt/mināt* ‘child’ in Central Kurdish is felicitous in environments where *baccaa* ‘child’ in Hindi is not (Dayal 2004: 406).

- (44) a. *mindāt la hamū shwen-èk yārī=y dá-kird.* Silemānī
 child in every place-INDF play=3SG PROG-did
 ‘Children were playing everywhere.’
 b. *mināt la harche jega bāzī=y á-kird.* Ardaḡānī
 child in every place play=3SG PROG-did
 ‘Children were playing everywhere.’
 (45) a. *caroN taraf bacce khel rabe the.* Hindi
 four ways child were.playing
 ‘Children (different ones) were playing everywhere.’
 b. # *caroN taraf baccaa khel rahaa thaa.*
 four ways child was.playing
 ‘The same child was playing everywhere.’

Curiously, *mindāt/mināt* can even appear with plural agreement morphology, as in (46), despite the lack of number marking on the noun itself.

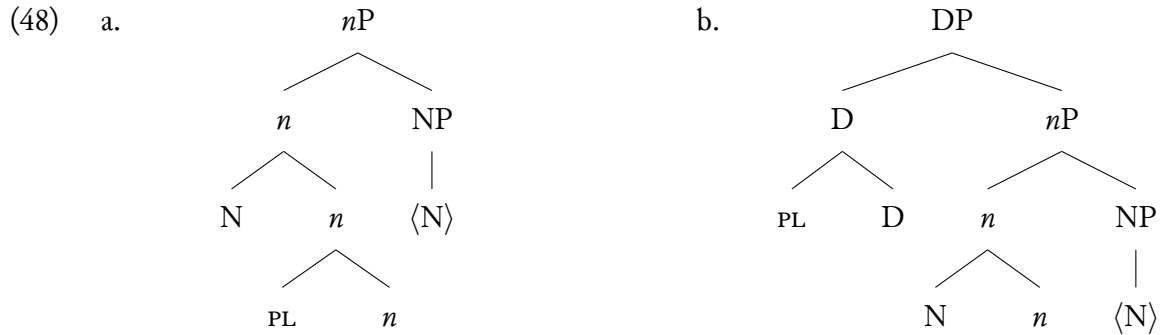
- (46) a. *mindāt la hamū shwen-èk yārī=yān dá-kird.* Silemānī
 child in every place-INDF play=3PL PROG-did
 ‘Children were playing everywhere.’
 b. *mināt la harche jega bāzī=yān á-kird.* Ardaḡānī
 child in every place play=3PL PROG-did
 ‘Children were playing everywhere.’

Even in Persian, which patterns like Central Kurdish in allowing a different-individual reading of *bache* ‘child’ in (47), the verb must be singular in form.

- (47) *bachē tu har gushe-i bāzi mi-kard-(*ān).* Persian
 child in every corner-INDF play DUR-did-PL
 ‘Children were playing everywhere.’

Taken together, the Central Kurdish facts suggest that plural marking is indeed optional, at least in indefinite expressions, and may not have the same syntactic representation that it does in languages where plural marking is required.

In light of the previous discussion, we pursue an account of Central Kurdish where a PLURALIZER (PL) can modify different heads in the noun phrase (Wiltschko 2008, 2009; Butler 2011; Cyrino & Espinal 2020). Moreover, in line with a growing body of research, we assume that arguments can be smaller than DP even if the language in question has definiteness morphology (Masullo 1992, 1996; Chierchia 1998b; Li 1998; Dayal 2004; Dobrovie-Sorin et al. 2005; Danon 2006; Pereltsvaig 2006; de Swart & Zwarts 2009; Dayal 2011b; Jambrović 2025). With respect to Central Kurdish, we claim that bare and indefinite nouns are minimally *n*Ps, that definite nouns are minimally DPs, and that PL adjoins to the highest head, *n* or D, that is present in the structure. Accordingly, PL modifies *n* in indefinite constructions, as in (48a), and D in definite and demonstrative constructions, as in (48b).¹²



Assuming that the obligatory raising of N to *n* precedes the optional adjunction of PL to *n*, this approach to number morphology derives both of the indefinite expressions in (29) above, which are repeated in (49a) and (49b) and have the structure in (49c).¹³

- (49) a. *zimānnās-ān-èk-ī ganj* Silemānī
 linguist-PL-INDF-IZ young
 ‘(some) young linguists’

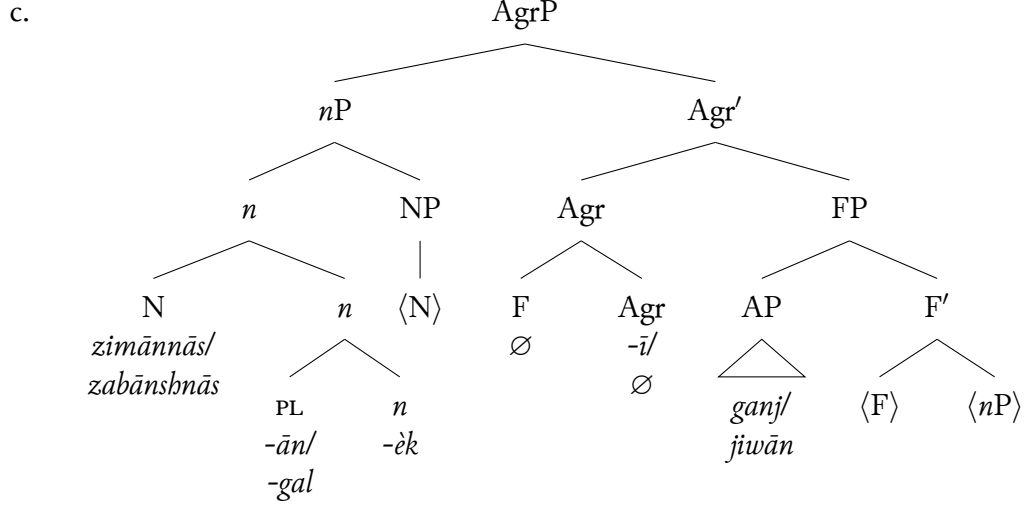
¹²It bears mentioning that PL combining with D does not face the same issue of compositionality that Num combining with DP does in (34) above. For instance, Mirrazi (2021: 674) proposes the following lexical entry for the suffix *-hā* in Persian, where the iota operator (ι) contributes uniqueness and $|x| > 1$ contributes plurality.

(i) $\llbracket -hā \rrbracket = \lambda P_{\langle e, t \rangle} . \iota x_e . P(x) \wedge |x| > 1$, if there exists a maximal plural entity in P , undefined otherwise.

¹³As for the cases in Ardaḡānī Kurdish where the indefinite suffix is the final element, such as *zabānshnāsgal jiwānèk* ‘(some) young linguists’ in (7), the morpheme order and interpretation suggest that *-èk* occupies a position above the domain of modification, such as QP. We leave a closer examination of this construction to future research.

- b. *zabānshnās-gal-èk-∅ jiwān*
 linguist-PL-INDF-IZ young
 ‘(some) young linguists’

Ardaḩānī



Before we present our analysis of definite and demonstrative noun phrases, which involve a DP layer, more needs to be said regarding the notion of uniqueness versus anaphoricity and how this distinction is expressed in Central Kurdish.

5. Decomposing *-aká*

Our focus in this section is the definite suffix *-aká* and the possibility that it is bimorphemic.¹⁴ We start by observing that demonstrative expressions in many Western Iranian languages reflect the template in (50), where the noun is enveloped by a demonstrative element and a definite suffix (MacKenzie 1961, Abdulla & McCarus 1967, McCarus 1997, Blau 2000, Thackston 2006, McCarus 2007, Holmberg & Odden 2008, McCarus 2009, Kahnemuyipour 2014, Öpengin 2016, Tahir 2018, Haig 2019b, Kalami 2020, Taghipour 2021, Ariyae 2023, Jambrović & Hassan 2023, Kalami 2023, Karim 2023, Peters et al. 2023, Shabani & Taghipour 2023).

(50) Dem N-D

There is only one type of demonstrative noun phrase that seems to deviate from this template, which is the plural anaphoric construction in the Ardaḩānī variety. In (51a) through (51c), the final element is *-á*, but in (51d), it is *-ān*.

- (51) a. {*am* / *aw*} *zabānshnās-á* Ardaḩānī
 PROX DIST linguist-DEF
 ‘this/that linguist’ (unique)

¹⁴Haig (2019a: 136–137) remarks that *-ak* likely evolved from *-ag*, a productive diminutive suffix in Middle Iranian, and became grammaticalized along the following path: “endearment > familiarity > identifiability, definiteness”. Even if *-ak* has diminutive origins, it would be difficult to maintain a low structural position for this suffix considering that syntactically complex APs can intervene between the noun and *-ak*, as shown in (13) and (14).

- b. {*am* / *aw*} *zabānshnās-gal-á*
PROX DIST linguist-PL-DEF
‘these/those linguists’ (unique)
- c. *aw zabānshnās-aká*
DIST linguist-DEF
‘that linguist’ (anaphoric)
- d. *aw zabānshnās-ak-ān*
DIST linguist-DEF-PL
‘those linguists’ (anaphoric)

Given that *-ān* only appears in the context of *-ak* in Ardaḡānī Kurdish, we argue that *-ān* is the plural counterpart of *-á*, as indicated by the revised segmentation and glossing of *aw zabānsh-nāsaká* ‘that linguist’ and *aw zabānshnāsakān* ‘those linguists’ in (52).

- (52) a. *aw zabānshnās-ak-á* Ardaḡānī
DIST linguist-*ak*-DEF
‘that linguist’ (anaphoric)
- b. *aw zabānshnās-ak-ān*
DIST linguist-*ak*-DEF.PL
‘those linguists’ (anaphoric)

We now motivate our claim that *-ak* and *-á* express separate aspects of definiteness in both varieties of Central Kurdish.

In the years since Ebert’s (1971a,b) seminal work on Fering, a variety of Frisian, many typologically diverse languages have been noted to make a morphological distinction between anaphoricity and uniqueness, or strong and weak definiteness (Hartmann 1982, Puig Waldmüller 2008, Wespel 2008, Schwarz 2009, Arkoh & Matthewson 2013, Schwarz 2013, Wiltschko 2013, Jenks 2015, Ingason 2016, Jenks 2018, Schwarz 2019, Šereikaitė 2019, Jenks & Konate 2022, Dawson & Jenks 2023, Dayal & Jiang 2023). In some of these languages, the difference is between two overt elements, and in others, it is between an element and its absence. We now demonstrate that there seems to be a three-way contrast in Central Kurdish in that nouns ending in *-aká*, nouns ending in *-á*, and bare nouns can all be interpreted as definite.

In prototypical anaphoric environments like (53) and (54), *-ak* cannot be excluded, suggesting that this morpheme encodes strong definiteness.

- (53) a. *kich-èk u kuř-èk la darawa yāri=yān dá-kird.* Silemānī
girl-INDF and boy-INDF outside play=3PL PROG-did
‘A girl and a boy were playing outside.’
- b. *kich-#(ak)-á pe-dá-kanī.*
girl-*ak*-DEF PVB-PROG-laughed.3SG
‘The girl was laughing.’
- (54) a. *kanishik-èk u koř-èk la daraw bāzī=yān á-kird.* Ardaḡānī
girl-INDF and boy-INDF outside play=3PL PROG-did
‘A girl and a boy were playing outside.’

- b. *kanishik-#(ak)-á á-kanya.*
 girl-*ak*-DEF PROG-laughed.3SG
 ‘The girl was laughing.’

For this reason, we gloss *-ak* as ANAPH in the remainder of this chapter. The question is whether there is sufficient evidence to classify *-á* in terms of weak definiteness.

If Central Kurdish were like Fering and other languages that have strong and weak definiteness morphology, one would expect *-á* to be used for situational uniqueness, but this suffix cannot be included in (55) or (56).

- (55) a. *khor-#(á) batde.* Silemānī
 sun-DEF rises
 ‘The sun is rising.’
- b. *sarok-#(á) l’era=ya.*
 boss-DEF in.here=is
 ‘The boss is here.’
- (56) a. *khwar-#(á) batte.* Ardafānī
 sun-DEF rises
 ‘The sun is rising.’
- b. *raīs-#(á) ha l’era.*
 boss-DEF is in.here
 ‘The boss is here.’

At the same time, *-á* is found to the exclusion of *-ak* “in narrative with known characters or things” (MacKenzie 1961: 51). For instance, the second occurrence of *zhin* ‘woman’ in (57) is marked solely with *-á* (Tahir 2018: 100).

- (57) a. *zhin-èk rā=y dá-kird baraw ema lagat dū mindāt-aká=y.* Bingird
 woman-INDF way=3SG PROG-made towards us with two child-DEF=3SG
 ‘A woman was running towards us with her two children.’
- b. *zhin-á bāwār=ī dá-kird bo yārmātī.*
 woman-DEF shout=3SG PROG-made for help
 ‘The woman was calling for help.’

It is telling that Fering and German both employ the weak definite article with referents that have been established as central figures of a narrative (Ebert 1971a,b; Schwarz 2009, 2019). There are, consequently, at least some empirical grounds to treat *-á* as a weak definite suffix.

Based on anecdotes and folktales that have been transcribed by MacKenzie (1962) and Öpengin (2016), Haig & Mohammadirad (2019) observe that speakers of Central Kurdish may omit definiteness marking altogether when referring to a principal character. For example, the narrator of the story “Who Deserves the Woman?” uses the indefinite suffix *èk* with the first mention of *najār* ‘carpenter’ in (58a), the definite suffix *-aká* with the second mention in (58b), and no suffix at all with the third mention in (58c) or any others (Öpengin 2016: 183–187).

- (58) Mukrī
- a. *najār-èk=ish=ī* *la bû.* HF.07
 carpenter-INDF=ADD=3SG in was
 ‘A carpenter was also there.’
- b. *najār-aká=sh* *kutî...* HF.08
 carpenter-DEF=ADD said.3SG
 ‘The carpenter said...’
- c. *najār* *nora=y* *bât.* HF.42
 carpenter turn=3SG.POSS came.3SG
 ‘The carpenter’s turn came.’

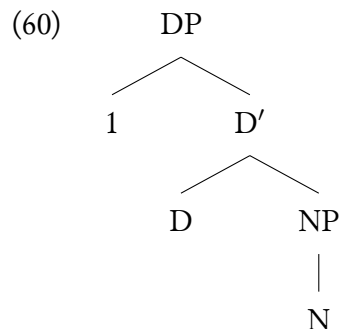
These authors liken such bare nouns to proper names, a comparison that Cheng & Sybesma (1999: 524) also make regarding Cantonese and the bare instance of *sin-saang* ‘teacher’ in (59a) versus the nonbare instance in (59b).

- (59) a. *Sin-saang mou lei.* Cantonese
 teacher not.have come
 ‘Teacher did not come.’
- b. *Go sin-saang mou lei.*
 CLF teacher not.have come
 ‘The teacher did not come.’

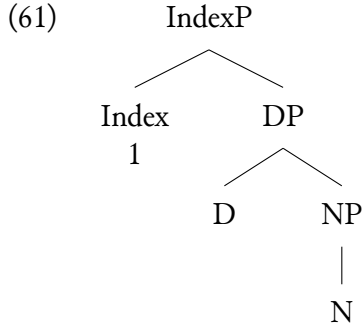
One could attribute the bare nouns *khôr/kbwar* ‘sun’ and *sarok/raïs* ‘boss’ in (55) and (56) above to the same phenomenon, in which case the absence of definiteness morphology is less surprising. Far more investigation is needed, but this initial foray into the relative distribution of nouns that have definite readings in Central Kurdish supports our proposal that *-ak* is a marker of anaphoricity and *-á* is a marker of uniqueness.

6. IndexP, DP, and fusion

When it comes to the structural representation of strong versus weak definiteness, there are two main approaches in the literature, both of which use an index for anaphoricity and associate D with uniqueness. Some authors position the index in the specifier of DP, as in (60), in which case there is a single determiner layer (Schwarz 2009; Jenks 2015, 2018; Schwarz 2019; Jenks & Konate 2022).

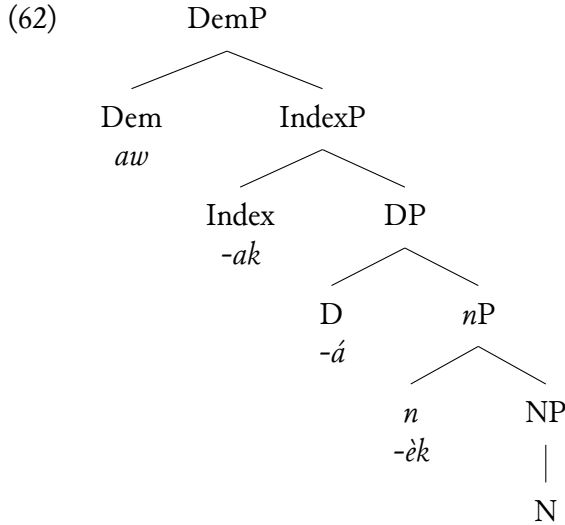


Other authors assign the index to a separate functional projection above DP, as in (61), which we label as IndexP for clarity (Cheng et al. 2017; Patel-Grosz & Grosz 2017; Bernstein et al. 2021, 2025).



We adopt the second configuration in our account of Central Kurdish because the specifiers of DP and IndexP are both needed as landing sites, though we assume that IndexP only projects in anaphoric environments based on general principles of economy (Chierchia 1998b, Cardinaletti & Starke 1999, Dayal 2004, Schlenker 2005, Dayal 2011a, Katzir 2011, Cheng et al. 2017, Patel-Grosz & Grosz 2017). Furthermore, we concur with Karimi (2007) and Tahir (2018) that D has an EPP-like requirement in Central Kurdish, one that is satisfied by movement of the nominal complex to the specifier of DP, and we extend the same condition to Index.

The architecture of the noun phrase that we propose for Central Kurdish is provided in (62), where we assign demonstrative elements like *aw* ‘that’ to Dem, the anaphoric suffix *-ak* to Index, the uniqueness suffix *-á* to D, and the indefinite suffix *-èk* to *n*.

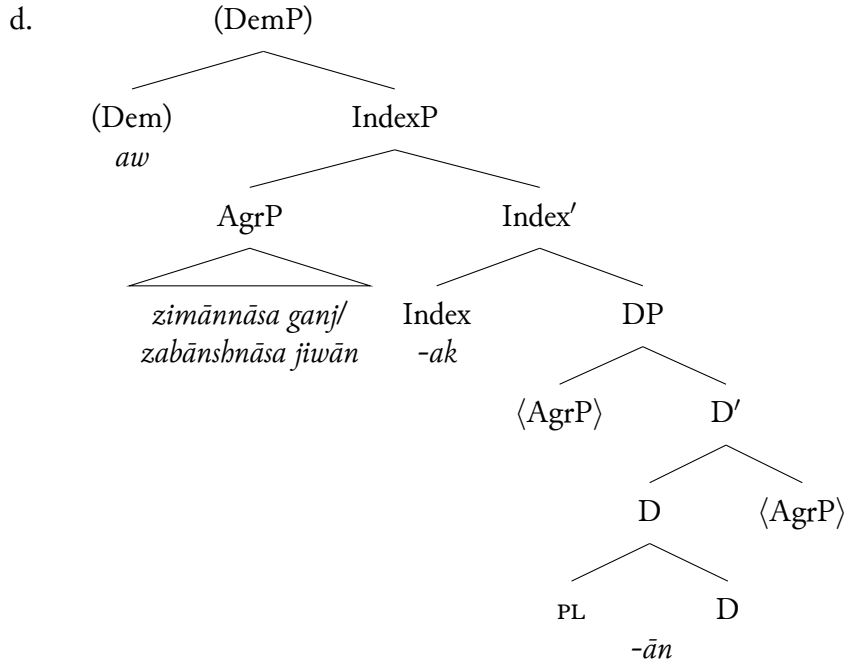


Since Tahir (2018) posits two determiner layers in Central Kurdish for independent reasons, as discussed in section 4, it is important to address the differences between our analyses. First, we model number morphology as a PLURALIZER (PL) that adjoins to D if DP projects and to *n* otherwise, not as a head that takes DP as its complement. Second, we decompose *-aká* into *-ak* and *-á* and locate these morphemes in Index and D, respectively, capturing the generalization that strong definiteness involves more functional material than weak definiteness. Third, we

maintain a low position for *-ək*, whereas Tahir situates the indefinite suffix in the higher of the two DPs in (34) above.

Beyond the structure in (62), only one further claim is needed to derive the morpheme order of definite and demonstrative noun phrases in Central Kurdish, one that is inspired by Taghipour's (2019, 2021) account of number and definiteness in Laki and Kalami's (2020) application of these insights to Ardaḡānī Kurdish: in the presence of Index, or *-ak*, PL and D undergo fusion and are spelled out together as *-ān*.¹⁵ For example, the anaphoric constructions in (63a) through (63c) all have the structure in (63d).

- (63) a. *zimānnās-a ganj-ak-ān* Silemānī
 linguist-IZ young-ANAPH-DEF.PL
 'the young linguists'
- b. *zabānshnās-a jiwān-ak-ān* Ardaḡānī
 linguist-IZ young-ANAPH-DEF.PL
 'the young linguists'
- c. *aw zabānshnās-a jiwān-ak-ān* Ardaḡānī
 DIST linguist-IZ young-ANAPH-DEF.PL
 'those young linguists' (anaphoric)

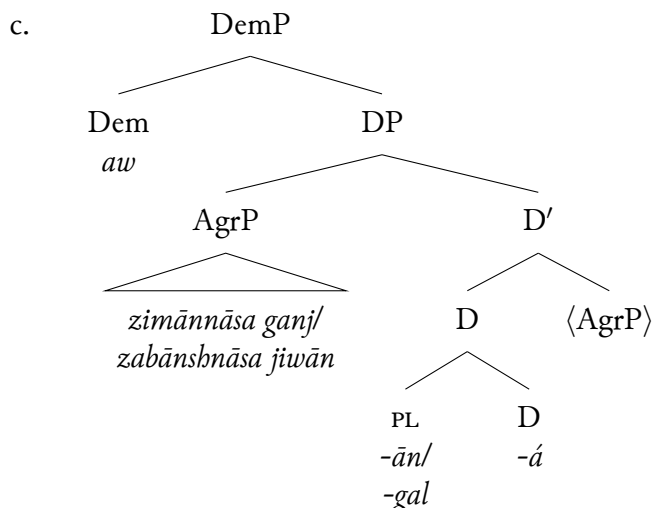


If, as we maintain, Index is the trigger for fusion, the expectation is that PL and D are realized separately in expressions that lack *-ak*. This prediction is borne out in (64), where *-ān/-gal* immediately precedes *-á*.

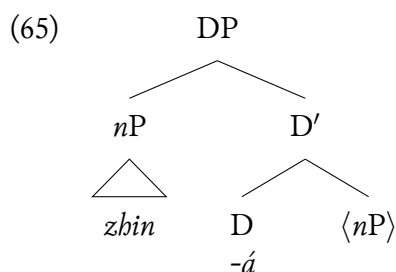
¹⁵We assume, as these authors do, that fusion is a postsyntactic operation (Halle & Marantz 1993, 1994; Harley & Noyer 1999). However, we do not resort to an [ANAPHORIC] feature on D or to Num-to-D raising.

- (64) a. *aw zimānnās-a ganj-ān-á* Silemānī
 DIST linguist-IZ young-PL-DEF
 ‘those young linguists’

- b. *aw zabānshnās-a jiwān-gal-á* Ardaḡānī
 DIST linguist-IZ young-PL-DEF
 ‘those young linguists’ (unique)



The only difference between the two varieties, which is superficial, is that PL and fused PL+D have the same form, *-ān*, in Silemānī Kurdish but distinct forms, *-gal* and *-ān*, in Ardaḡānī Kurdish. Finally, definite nouns that are marked with *-á* rather than *-aká*, such as *zhiná* ‘the woman’ in (57b) above, have the representation in (65).



In summary, all that is required to capture the various definite and demonstrative constructions in Central Kurdish is to decompose *-aká/-akān* into *-ak* and *-á/-ān* and posit fusion of PL and D in the environment of Index.

In section 4, we showed that bare singular nouns in Central Kurdish permit number-neutral readings and exhibit flexible subject-verb agreement. The definite singular nouns in (66), on the other hand, do not.

- (66) a. *mindāt-ak-á la darawa yārī={y / *yān} dā-kird.* Silemānī
 child-ANAPH-DEF in outside play=3SG 3PL PROG-did
 ‘The child was playing outside.’
 Unavailable: ‘The children were playing outside.’

- b. *mināt-ak-á la daraw bāzī={y / *yān} á-kird.* Ardaḩānī
 child-ANAPH-DEF in outside play=3SG 3PL PROG-did
 ‘The child was playing outside.’
 Unavailable: ‘The children were playing outside.’

In these cases, the referent is known, so one could attribute the unavailability of a plural reading to a principle like Heim’s (1991) Maximize Presupposition and the notion that singular *mindātaká/minātaká* ‘the child’ competes with plural *mindātakān/minātakān* ‘the children’. Given this possibility, we do not consider this asymmetry between bare nouns and definite nouns to undermine the treatment of number morphology that we have developed in this chapter.

7. Conclusion

In Central Kurdish, there are two morpheme orders that are not readily explained by standard assumptions regarding the structure of the noun phrase and movement within it: N AP D Num and N AP Num D. The fact that roll-up movement takes place in this language makes the positioning of modifiers inside of number morphology all the more puzzling. To account for the variable position of plural marking in Silemānī and Ardaḩānī Kurdish, we proposed that an optional PLURALIZER adjoins to D when DP projects and to *n* otherwise. We then argued that the definite suffix *-aká* is bimorphemic, where *-ak* conveys anaphoricity and *-á* conveys uniqueness. Finally, we assigned *-ak* to IndexP and *-á* to DP and maintained that PL and D undergo fusion in the context of Index, in which case these nodes are spelled out together as *-ān* in both the Silemānī and Ardaḩānī varieties.

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Abbreviations

1 = first person, 2 = second person, 3 = third person, ADD = additive, ANAPH = anaphoric, CLF = classifier, CMPR = comparative, DEF = definite, DIST = distal, DUR = durative, INDF = indefinite, IZ = *izāfa*, OBL = oblique, PL = plural, POSS = possessive, PROG = progressive, PROX = proximal, PVB = preverb, SG = singular, SUP = superlative

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